BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: DIK-HSSC-XII-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2021 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. The unit of mutual inductance is _____

(A) Volt
(B) Ampare
(C) Henry
(D) Tesla

Qii. In transistor which junction is forward based _____

(A) Collector
(B) Emitter
(C) Both A and B
(D) Sometime A & Sometime B

Qiii. Unit of decay constant λ is _____

(A) m.s
(B) m^{-1}
(C) m
(D) s^{-1}

Qiv. One e.m.u is equal to _____

(A) 0.51 MeV
(B) 931 MeV
(C) 238 MeV
(D) 231 MeV

Qv. _____ radiation ionized air molecules strongly.

(A) Alpha
(B) Beta
(C) Gama
(D) Both B&C

Qvi. $1NA^{-1}m$ is equal to _____

(A) 1 Henry
(B) 1 Tesla
(C) 1 Becquerel
(D) 1 Gauss

Qvii. For step down transformer _____

(A) Vs > Vp
(B) Vs ≥ Vp
(C) Vs < Vp
(D) Vs = Vp

Qviii. In case of DC the capacitive resistance Xc has_____ value.

(A) Finite
(B) Zero
(C) Infinite
(D) Both A&B

Qix. A size of nucleus is equal to _____

(A) 10^{-14}m
(B) 10^{-10}m
(C) 10^{-11}m
(D) 10^{-12}m

Qx. Rectification is done by _____

(A) Choak
(B) Diode
(C) Transformer
(D) Inductor

Qxi. _____ is the example of Ferro-magnetic material.

(A) Copper
(B) Antimony
(C) Zinc
(D) Cobalt

Qxii. 1.602 x 10^{-19} Joules is equal to _____

(A) 1eV
(B) 1V
(C) 1GeV
(D) 1MeV

Qxiii. Dimension of plank constant is equivalent to_____

(A) Torque
(B) Momentum
(C) Angular momentum
(D) Energy

Qxiv. Ampere law is applicable to _____ path.

(A) Any closed
(B) Circular
(C) Rectangular
(D) None

Qxv. One weber is equal to _____

(A) $N.A^2/m$
(B) $N.m^2/A$
(C) N.A/m
(D) N.m/A

Qxvi. Electric potential of earth is taken to be zero because earth is a good _____

(A) Semi-conductor
(B) Conductor
(C) Insulator
(D) di-electric

Qxvii. The algebraic sum of potential changes in a closed circuit is _____

(A) Potential difference
(B) Resistance
(C) Zero
(D) None of these

Qxviii. The magnetic field around a current carrying wire is _____

(A) BαI
(B) BαI/R₀ [illegible in source scan]
(C) BαI/R [illegible in source scan]
(D) Both A&B

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 40

Q2. Attempt any Ten questions of the following. All questions carry equal marks

i. Why Bohar extends quantum theory to the structure of atom?

ii. What is the magnitude of force on a charge a moving with a velocity V in a magnetic field B?

iii. How a 5mA, 100Ω galvanometer into 20V volt meter?

iv. The rms value of current in AC circuit is 10A. What is the peak current?

v. Describe the four basic forces of nature.

vi. Why a rise in temperature of a conductor is accompanied by a rise in the resistance?

vii. How mass defect and binding energy related to a nucleus?

viii. Discuss the classification of magnetic materials.

ix. Radium has a half-life of about 1600 years if the universe was formed 5 billion or more year ago why is there any radium left now?

x. What is super conductor?

xi. Why does the spectrum of hydrogen consist of many lines even through a hydrogen atom has only a single electron?

xii. What is meant by radioactivity?

xiii. Explain law of radioactive decay.

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any three Question. Each question carries equal marks.

Q.3 (a) State and explain Gauss's law for electro statics.

(b) Show that the electric field intensity due to infinite sheet of positive charge is $E = \delta/(2\varepsilon_0)$

Q.4 (a) Define, explain and write a brief note on Electric Transformer.

(b) How a galvanometer is converted into ammeter.

Q.5 (a) What is De-Broglie wave? Derive the expression for De-Broglie wave length $\lambda=h/mv$

b. Find the De-Broglie wave length of electron moving with a speed of 200 m/s.

Q.6 Write short notes on any two of the following: a. Rheostat b. AC through resistor c. Energy Band Theory